

DERIVATIVES 101 • DECK 0

# Foundations

The handful of building blocks that every product on this desk is made from.  
No prior finance knowledge assumed.

# Six decks, one idea repeated

0

## Foundations

Spot, options, premium, volatility, barriers, coupons, autocalls, baskets. The vocabulary everything else reuses.

1

## KODA / KODD

Accumulators and decumulators. Buy a little every day at a discount — until the market moves against you.

2

## Reverse convertibles

Equity and FX. A high coupon against your principal. Conditional coupons, callable, autocallable, lock-in.

3

## Asian FCN & ELN

The short-dated, worst-of, physically-settled notes that dominate Hong Kong, Singapore and Taiwan flow.

4

## Discount & bonus

The European listed side. Premium taken as a price discount or a conditional bonus instead of a coupon.

5

## Protection & leverage

Capital protected notes at one end, mini futures at the other. Same issuer credit underneath both.

Every product in decks 1–5 is the same trade wearing a different suit: the client sells optionality and gets paid for it.

# Four investment families, plus leverage

|                                                          |                                                                          |                                                                  |                                                             |                                                        |
|----------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------|
| 11xx                                                     | 12xx                                                                     | 13xx                                                             | 14xx                                                        | 2xxx                                                   |
| <b>Capital protection</b>                                | <b>Yield enhancement</b>                                                 | <b>Participation</b>                                             | <b>Additional credit risk</b>                               | <b>Leverage</b>                                        |
| The floor is bought first; whatever is left buys upside. | Capped upside in exchange for a known return. The biggest family by far. | Open-ended upside. Trackers, outperformance, bonus certificates. | A second credit exposure bolted on top of the issuer's own. | Warrants, knock-outs, mini futures, constant leverage. |
| Deck 5                                                   | Decks 1-4                                                                | Deck 4                                                           | Not covered                                                 | Deck 5                                                 |

**The family is decided by the shape of the return, not by how the product is built. Add a cap to a bonus certificate and it moves from participation to yield enhancement, even though only one option leg changed.**

Two traps: the Swiss map has no code for a capped bonus, and 1260 means different products in Switzerland and the rest of Europe.

# Structured products are recipes, not ingredients

- A structured product is a bundle. Unwrap it and you find a deposit or bond, plus one or more options.
- The client is almost always a net seller of options — that is where the headline yield or discount comes from.
- The bank is the buyer of those options, and then spends the product's life hedging what it bought.
- So there are only two questions worth asking about any new payoff: what has the client sold, and what makes it worse?

## A yield-enhancement note =

### Zero-coupon bond

Repays 100 at maturity. Costs less than 100 today — the gap is your budget.

### + option premium received

The client sells a put (or a strip of them). The premium buys the coupon.

### – issuer margin & hedging cost

What the desk keeps, and what it expects to pay to stay hedged.

### = the coupon on the term sheet

Everything the client sees is the residual of the three lines above.

# Spot, forward, and why the forward is not a forecast

## Spot

The price you pay to own the asset right now. HSBC at HK\$95.40. USD/JPY at 147.20.

## Forward

The price agreed today for delivery on a future date. Fixed now, settled later.

## Carry

Forward = spot adjusted for interest earned, funding paid, and dividends foregone. Nothing else.

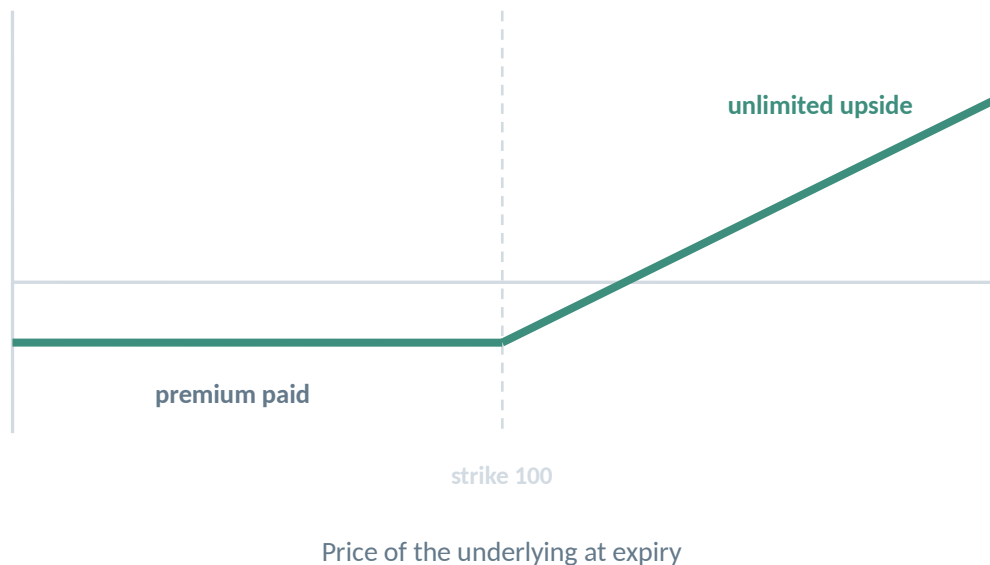
**A one-year forward on a stock trades below spot when the dividend yield exceeds the funding rate. That is arithmetic, not a view.**

This matters because clients regularly read a 'discount to spot' on a term sheet as a bargain, when part of it is simply carry. The rest — the part that actually pays for the product — is option premium.

## BUILDING BLOCK 2

# A right for one side, an obligation for the other

Long call — the right to buy at 100



Short put — the obligation to buy at 100



Almost every product we trade puts the client on the right-hand chart. A small capped gain, paid up front, against a loss that runs all the way to zero.

# Premium is the price of uncertainty

- **Volatility** measures how much the underlying moves, not which way.
- Realised volatility is what actually happened. Implied volatility is what the option market is charging for what might happen.
- Higher implied vol means a bigger premium — so a scared market funds a fatter coupon for the same risk barrier.
- That is why coupons on the same structure jumped sharply in April 2025 when the VIX spiked, without a single term on the sheet changing.

## Vol ↑ → Premium ↑

More movement priced in, so the option the client sells is worth more.

## Tenor ↑ → Premium ↑

More time, more chance of a big move — but also more time to be wrong.

## Strike ↓ → Premium ↓

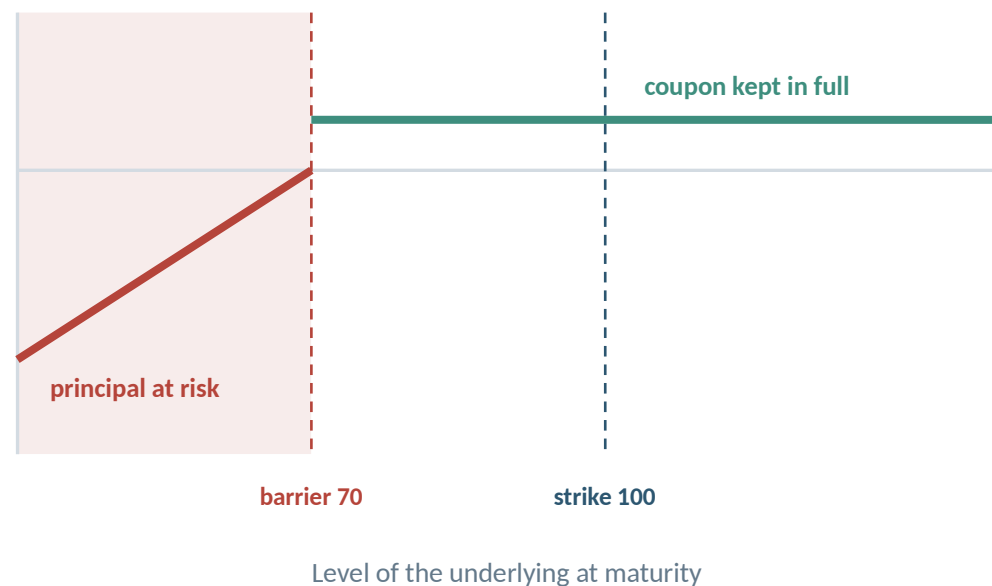
A put struck further below spot is less likely to be exercised, so it pays less.

## Assets ↑ → Premium ↑

A worst-of basket sells more optionality, so the coupon rises with each name added.

# Two charts tell you almost everything

Payoff diagram — what you get, given where it ends



Path chart — whether the route mattered



**If the product has a barrier, the payoff diagram alone is a lie. You have to know whether the barrier was touched.**



# Barriers: the same product, two different lives

## Knock-out (KO)

Above spot, usually. Touch it and the trade ends early — often profitably for the client, but the profit is capped and the trade is gone.

*Also called the autocall or early-redemption trigger.*

## Knock-in (KI)

Below spot. Touch it and the client's downside switches on. Until then the note behaves like a bond; afterwards it behaves like the stock.

*Hong Kong retail ELIs call the protection above this level an 'airbag'.*

## Observation style changes the price more than most people expect

### European

Observed once, at maturity. Only the final close matters.

### American / continuous

Observed every tick of every day. The low of the year is what counts.

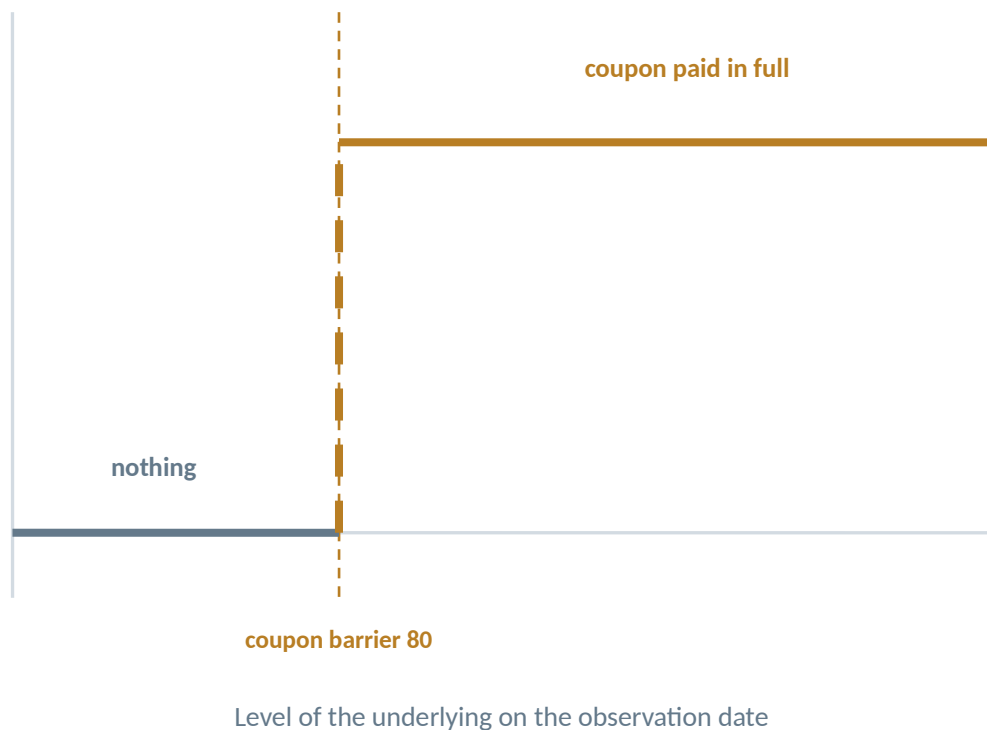
### Discrete / daily close

Observed on scheduled dates only — daily, monthly or at each coupon date.

*The market has drifted from continuous towards European knock-ins since 2015 — a quiet but material de-risking of the whole product set.*

# Coupons: fixed, conditional, or remembered

A conditional coupon is a digital option



## Fixed / unconditional

Paid whatever the underlying does. The defining feature of a Fixed Coupon Note.

## Conditional

Paid only if the underlying is above a coupon barrier on the observation date.

## Memory

Missed coupons are not lost — they are banked and paid later if the barrier is regained.

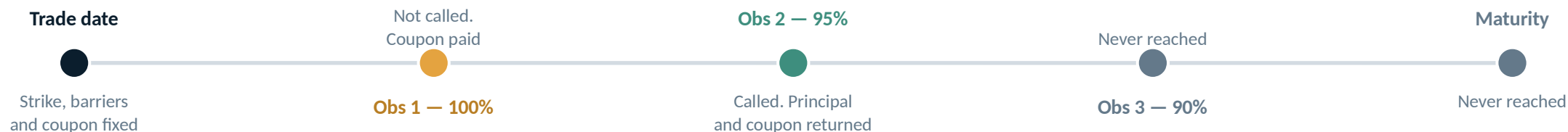
## Accrual

Accrues only on the days the underlying closes above a level. Part-paid coupons are normal.

*A coupon that is 'guaranteed' is guaranteed in isolation. It is never a statement about the principal.*

# Autocall: the product decides when it ends, not you

A three-year note that usually ends long before three years



## Step-down triggers

Each observation calls at or below the previous level, so the note gets easier to redeem as it ages. 85 / 85 / 85 / 80 / 75 / 70 is a normal Korean schedule.

## Non-call period

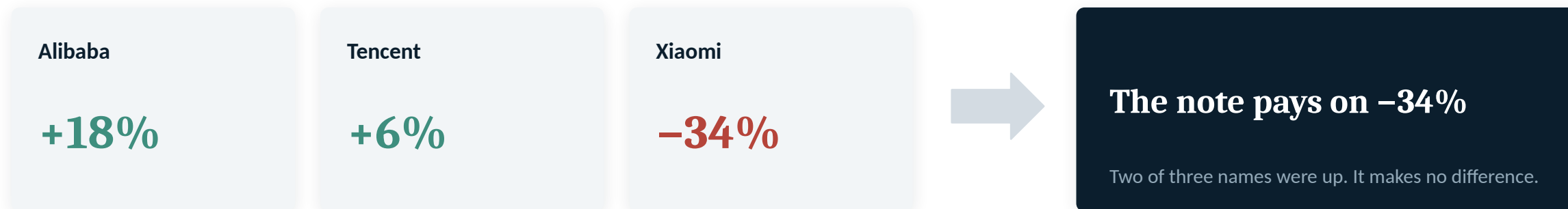
Three to six months at the start where no call can happen. It guarantees the client at least one or two coupons.

## Why it matters to us

Expected life is not contractual life. Cashflow schedules, accruals and position ageing all have to handle early termination as the normal case, not the exception.

*Autocall probability is why a client who wants three years of coupons rarely gets them: a rising market redeems the note early.*

# Worst-of: the basket that only remembers its weakest name



- More names means a higher coupon — and a strictly worse expected outcome.
- Adding a fourth name cannot improve the worst performer. It can only make it worse. The extra coupon is the price of that.
- The extra coupon is paid for with correlation. A worst-of option is worth most when the names decouple, because the worst performer is then further from the pack — so the client is selling dispersion without ever being told so.
- Two to four names is the market norm. Three is the most common structure we see.

# Three levers, and you cannot pull all three

## 1 Coupon

How much the client is paid

*Raise it and something else must give.*

## 2 Protection

How far the market can fall before it hurts

*A deeper barrier costs coupon.*

## 3 Participation

How much upside the client keeps

*Most of these products keep none.*

**When a client asks for a higher coupon, they are asking for a worse barrier, a shorter fuse, or another name in the basket. There is no free version.**

This is the single most useful sentence to have in your head when reading a term sheet. Every unusual number on the page is paid for by another number on the same page.

# The desk buys the option, then spends a year defending it

## We are long the downside option

The client sold it to us. On day one that is a comfortable place to be.

## We are short the coupon

Paid whether or not the hedge performs. It is a fixed liability against a moving asset.

## The hedge is not static

As spot falls, our sensitivity to volatility rises, peaks, and then collapses — forcing us to buy volatility back at the worst possible moment.

## Everyone flips at once

The whole street issues on the same names at similar levels. When the hedge turns, it turns for all of us simultaneously, in the same direction.

*Deck 2 covers this properly. For now: the client's risk and the desk's risk are not mirror images — they fail in different ways, at different times.*

# The anatomy of a termsheet

## 1 Classification line

The SSPA / EUSIPA category and code, plus the settlement and headline feature. Read this before anything else — it tells you which family you are in.

## 2 Distribution restriction

Who may be sold this, under which regime. In Switzerland this is the FinSA line, and it is the reason the same payoff comes in different wrappers.

## 3 Terms

Issue price, denomination, issue size, coupon, maximum yield, tenor, and the four dates. Everything commercial is in this block.

## 4 Underlying

Initial Level, Strike, Barrier, Trigger Barrier, Payout Threshold, Ratio, Valuation Time. Every level is quoted twice — absolute and as a percentage of initial.

## 5 Redemption

The payoff, written as nested conditions. Final Redemption Amount, then Barrier Event, then the observation dates and period.

## 6 Early redemption & payout

Trigger Redemption for the autocall, Payout for the coupon. Each has its own barrier, its own observation dates and its own table.

**Risk factors come last in the document and matter first. Read backwards on your second pass.**

# Nine terms that decide the whole payoff

|                  |                                                                                                                                             |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Initial Level    | The reference fixed on the Initial Fixing Date. Everything else is quoted as a percentage of it.                                            |
| Strike           | The level the payoff is measured against, and the price at which shares are delivered.                                                      |
| Barrier          | Where principal protection is lost. Check whether it is observed continuously or only at the final fixing.                                  |
| Trigger Barrier  | The autocall level. A Trigger Event redeems the note early at the Trigger Redemption Price.                                                 |
| Payout Threshold | The level a conditional coupon needs. It is a separate number from the Barrier and can sit either side of it.                               |
| Ratio            | $\text{Denomination} \div \text{Strike}$ . The number of shares delivered on physical settlement, rounded down, with cash for the fraction. |
| Maximum Yield    | The most the product can pay. Compare it with the headline coupon before you believe the headline.                                          |
| Valuation Time   | Usually the scheduled closing time — but not always, and for FX often a named time in a named city.                                         |
| The four dates   | Initial Fixing, Issue, Final Fixing, Final Redemption. They are all different, and the gaps between them are real.                          |

**Two weeks can separate the Initial Fixing Date from the Issue Date. The trade is struck long before the money moves.**



# The words you will hear on day one

**Notional**

The face amount the payoff is calculated on. Not what the client paid.

**Strike**

The reference level for the option. Often set below spot.

**Knock-in**

The level below which the client's downside switches on.

**Knock-out / autocall**

The level at which the trade terminates early.

**Worst-of**

Payoff references the weakest performer in a basket.

**Gearing**

A multiplier applied to the loss side. Usually 2x.

**Fixing**

The scheduled observation of a price. Fixings drive everything.

**Physical settlement**

The client is delivered shares instead of cash.

**Vega**

Sensitivity to implied volatility.

**Skew**

The market charging more for downside options than upside ones.

**Tenor**

Contractual life. Rarely the actual life.

**PI / AI / SPI**

Professional, accredited and sophisticated investor categories.

# Five things, and then we start on real products

- 1 Unwrap the bundle. Bond plus options, always.
- 2 Find what the client sold. It is nearly always a put, or a strip of them.
- 3 Check whether the path matters. Barriers make the payoff diagram incomplete.
- 4 Assume early termination. Contractual tenor is the exception, not the rule.
- 5 Remember the trade-off. A better number on the term sheet was paid for by a worse one.